Measuring and Mitigating Social Biases in Lightweight Large Language Models

Giovanni Filippini[1*]

[1*]Department of Computer Science "Giovanni degli Antoni", University of Milan, Milan, Italy.

Corresponding author(s). E-mail(s): giovanni.filippini@studenti.unimi.it;

Abstract

The presence of social bias in large language models poses a significant challenge in the field of natural language processing, as it hinders the fairness and inclusivity of AI systems. The present study undertakes a systematic investigation into the presence of social biases in lightweight transformer models, with the objective of measuring and mitigating these biases. The research evaluates gender, racial, religious, and professional stereotypes in pre-trained models, specifically a model from the SmolLM2 family, using the StereoSet benchmark. The approach implemented involves the utilization of bias detection metrics and visualization techniques to quantify and qualify these biases. Subsequently, mitigation strategies are applied through fine-tuning on counter-stereotypical data. The findings are then juxtaposed with the initial models, underscoring the endurance of social biases within transformer models and the efficacy of computationally efficient mitigation techniques.

Keywords: Natural Language Processing, Social Bias, Large Language Models, Bias Mitigation, Fine-tuning, StereoSet

1 Introduction

Large language models have become foundational elements of contemporary natural language processing systems, providing the underlying functionality for a wide range of applications, including text generation and sentiment analysis. However, these models are trained on extensive corpora of human-written text, which inherently contain and potentially amplify societal biases related to gender, race, religion, and profession. When such biases are propagated through AI systems, they have the potential to perpetuate harmful stereotypes and lead to discriminatory outcomes in applications ranging from hiring algorithms to content recommendation systems.

Recent research has demonstrated that transformer-based language models, while exhibiting remarkable linguistic capabilities, are particularly susceptible to encoding and reproducing societal biases. Recent studies[1–3] have demonstrated that models, such as BERT, RoBERTa, and their distilled variants, have the capacity to associate specific professions with particular genders, ethnicities with negative attributes, and religions with stereotypical characteristics. These associations are not merely theoretical concerns, they have real-world implications when these models are deployed in production systems.

The present study focuses on the identification, measurement, and mitigation of social biases in lightweight language models. In particular, our research will deal with three primary forms of bias: gender, racial, and professional stereotypes. To conduct the research, a four-step methodology has been employed:

1. The preliminary step involves the selection and evaluation of pre-trained models;
2. The quantification of prevailing biases is achieved through the utilization of recognized standards;
3. The implementation of bias mitigation techniques is imperative;
4. The re-evaluation of the model to ascertain the extent of the improvements.

2 Methodology

2.1 Research Questions

The primary issue identified in the introduction will be addressed, and several foundational inquiries will be examined with the aim of elucidating the extent to which lightweight language models encode and reproduce social biases related to gender, race, religion, and profession. The investigation's objective is twofold: first, to identify the categories of bias that are most prevalent in these models, and second, to understand how these biases manifest in model predictions. Additionally, the research examines whether bias mitigation techniques can significantly reduce these biases without compromising model performance on downstream tasks.

2.2 Problem Formulation

The problem of bias in language models can be formally defined as measuring and mitigating unwanted associations between social groups and attributes that reinforce stereotypes.

For a language model M, given a set of social group terms G (e.g., gendered terms, racial identifiers, religious affiliations) and attribute terms A (e.g., professions, personality traits), the aim is to measure the association strength $S(g, a)$ between each $g \in G$ and $a \in A$.

A model exhibits bias when $S(g_1, a) \neq S(g_2, a)$ for stereotypical associations where fairness would dictate equal treatment, which means that the model is more likely to associate a certain group with a certain attribute.

This measurement is implemented through the utilization of the StereoSet[4] benchmark, which includes a series of intra-sentence and inter-sentence completion tasks, designed to identify stereotypical, anti-stereotypical, and unrelated associations. The benchmark returns a Stereotype Score (SS) that measures bias, and a Language Modeling Score (LMS) that measures task performance. These are combined into a Single Sentence Score (SSS) that balances fairness and utility:

$$SSS = LMS \times (1 - |SS - 0.5| \times 2)$$

A model with perfect fairness would have $SS = 0.5$ (equal probability of stereotypical and anti-stereotypical completions), while maintaining high LMS values.

2.3 Research Approach

The research approach follows a systematic methodology, beginning with the selection of lightweight, transformer-based pre-trained model suitable for efficient deployment and evaluation. The focus will be on SmolLM2[5] model, specifically the version with 360 million parameters (designated hereafter as SmolLM2-360M), developed by Hugging Face, which offers a good balance between performance and computational efficiency. The bias evaluation phase employs benchmarking each model across four critical bias domains: gender, race, religion, and profession. For each domain, quantitative metrics including the Stereotype Score, Language Modeling Score, and Single Sentence Score are calculated and complemented with qualitative analyses through word embedding projections and bias association heatmaps.

The bias mitigation strategy involves fine-tuning the model on counter-stereotypical examples extracted from the StereoSet benchmark, which enables the model to recognize and favor non-stereotypical associations while preserving core language understanding capabilities. The fine-tuning process utilizes optimized parameters with three epochs and a batch size of 16 to ensure both effectiveness and computational efficiency, and the final phase encompasses a comparative analysis where each model undergoes re-evaluation after fine-tuning, comparing pre- and post-intervention results across all metrics in order to examine in detail the trade-offs between bias reduction and model performance while visualizing changes through embedding projections and bias score distributions.

3 Implementation

3.1 Datasets

The StereoSet benchmark dataset, which was designed to assess stereotypical biases in language models, encompasses a diverse array of examples spanning the four distinct domains that have been previously defined:

- **Gender**: A total of 771 examples were collected, encompassing stereotypes related to gender identity and roles.
- **Profession**: A total of 2,398 examples pertaining to occupational stereotypes have been documented.

- **Race**: A total of 2,976 examples were identified, pertaining to racial and ethnic stereotypes.
- **Religion**: A total of 247 examples addressing religious stereotypes and associations.

The dataset is structured as a collection of context sentences with multiple completions: stereotypical, anti-stereotypical, and unrelated. For evaluation, the development split containing 6,392 examples is used to assess model biases. For fine-tuning, anti-stereotypical examples are specifically extracted from the test split, which provides a larger and more diverse set of examples for training.

In the codebase, the `StereoSetDataset` class handles dataset downloading, caching, and preprocessing, transforming the raw data into a format suitable for both evaluation and fine-tuning. The contained examples are organized by bias category in order to enable detailed domain-specific analysis.

3.2 Evaluation Metrics

The implementation incorporates a suite of metrics designed to quantify bias in language models, where the Stereotype Score serves as the primary metric, measuring a model's preference for stereotypical associations where an SS of 0.5 indicates no bias, values above 0.5 indicate stereotype bias, and values below 0.5 indicate anti-stereotype bias. Complementing this, the Bias Severity metric, calculated as $|SS-0.5|$, quantifies the magnitude of bias regardless of direction, with higher values indicating stronger bias. In order to detect whether the model tends toward stereotypical (+1) or anti-stereotypical (-1) associations, a Bias Direction is used to indicate this situation; while the Bias Difference provides the absolute difference between stereotype and anti-stereotype scores, providing a raw measure of preference strength. Finally, the Bias Ratio represents the ratio of probabilities assigned to stereotypical versus anti-stereotypical completions.

The `BiasEvaluator` class handles the evaluation process, automatically detecting whether the model is a masked language model, like BERT, or a causal language model, like SmoLM2, and applying the appropriate scoring mechanism. For masked language models, the masked token prediction loss[1] is used, while for causal language models, next-token prediction loss[2] across the entire sequence is employed.

To visualize bias patterns, three visualization techniques are implemented:

- **Bias distribution histograms**: shows the frequency of different bias severity levels.
- **Category-based bias radar charts**: compares bias across domains.
- **Heatmaps**: illustrates stereotype vs. anti-stereotype preference patterns.

3.3 Bias Mitigation Techniques

The bias mitigation approach deals with fine-tuning with counter-stereotypical examples. The data preparation phase involves extracting examples labeled as "anti-stereotype" from the StereoSet dataset, creating a curated training set that explicitly

[1] Measures how ably the model predicts missing (or "masked") words in a sentence.
[2] Measures how a language model predicts the next word at each position in a sequence.

contradicts common stereotypes. The `BiasMitigationDataset`, implemented as a custom PyTorch dataset class, handles tokenization and prepares input-label pairs for training with appropriate padding token management.

The core mitigation process utilizes the `ModelFineTuner` class, which implements fine-tuning with an AdamW optimizer[3] incorporating weight decay for regularization, a linear learning rate scheduler with optional warmup steps, and configurable training parameters including epochs, batch size, and learning rate. The system also includes checkpoint saving capability for model persistence.

This implementation maintains the original model architecture and the majority of the parameters, with the fine-tuning specifically focused on adjusting the model's token distributions to mitigate stereotypical associations, by also maintaining the model's general language understanding capabilities while improving fairness across bias categories.

4 Results

4.1 Pre-fine-tuning Results

The preliminary assessment of the SmolLM2-360M model revealed significant biases across all four domains. The overall Stereotype Score of 0.4915 indicates a slight anti-stereotype bias, meaning the model somewhat favors anti-stereotypical completions over stereotypical ones. Table 1 presents the detailed evaluation metrics for each bias category from the original model.

Table 1 Pre-fine-tuning Bias Metrics for SmolLM2-360M

Category	SS Score	Stereotype Score	Anti-Stereotype Score	Bias Severity	Bias Δ
Overall	0.4915	-4.7064	-4.8652	0.0085	0.1587
Gender	0.4884	-4.5414	-4.7585	0.0116	0.2171
Profession	0.4923	-4.9038	-5.0532	0.0077	0.1493
Race	0.4920	-4.6107	-4.7540	0.0080	0.1433
Religion	0.4863	-4.4585	-4.7117	0.0137	0.2532

Religion emerged as the most biased category with a bias severity of 0.0137, followed by gender bias (0.0116). The anti-stereotype direction across all categories (indicated by SS scores < 0.5) suggests that the model may have been trained with some bias awareness already incorporated. However, the ideal SS score would be 0.5, indicating true neutrality between stereotypical and anti-stereotypical associations.

The bias difference metrics showed the largest gaps in religion (0.2532) and gender (0.2171), confirming these domains as areas of particular concern, highlighting the persistent nature of social biases in transformer models, even in more recent lightweight architectures designed for efficient deployment.

[3] An optimization algorithm that decouples weight decay from the gradient-based update of weights, improving generalization in a model's training.

Figure 1 shows the bias distribution across different categories, visualizing the concentration of bias values across the model's predictions.

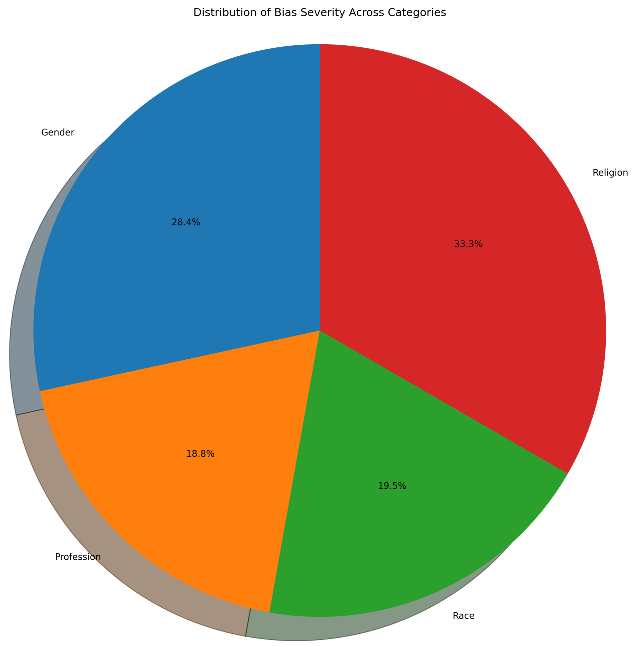

Fig. 1 Bias distribution for the original SmolLM2-360M model

The heatmap visualization in Figure 2 provides a more detailed view of stereotype and anti-stereotype associations, highlighting specific areas where bias is most pronounced.

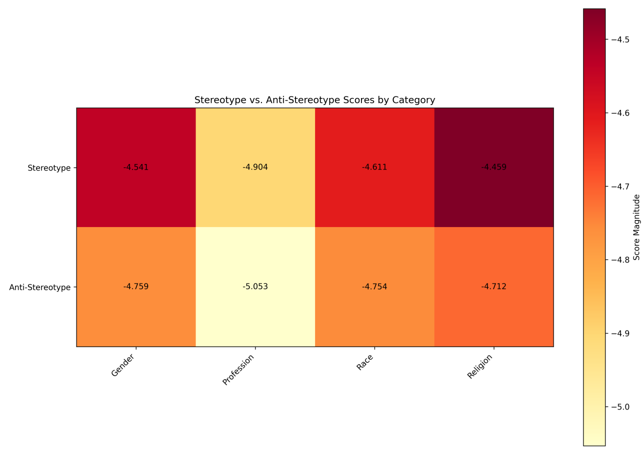

Fig. 2 Heatmap of stereotype vs. anti-stereotype associations in the original model

These data reveal that certain occupations, such as "nurse" and "teacher," showed stronger gender associations, while religious terms exhibited notable correlations with specific attributes and geographical regions. The bias visualization radar chart in the Figure 3 further illustrates the relative bias severity across the four domains.

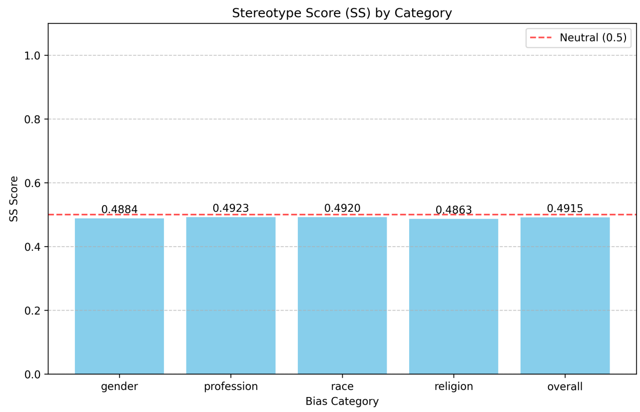

Fig. 3 Radar chart of bias severity across domains in the original model

The results highlight the persistent nature of social biases in transformer models, even in more recent lightweight architectures designed for efficient deployment.

4.2 Post-fine-tuning Results

After fine-tuning on counter-stereotypical examples, significant changes were observed in the bias metrics of the model, as we can see in detail in Table 2.

Table 2 Post-fine-tuning Bias Metrics for SmolLM2-360M

Category	SS Score	Stereotype Score	Anti-Stereotype Score	Count
Overall	0.5584	-2.5307	-2.0009	6392
Gender	0.5624	-2.7129	-2.1218	771
Profession	0.5582	-2.7175	-2.1534	2398
Race	0.5573	-2.3352	-1.8504	2976
Religion	0.5599	-2.5029	-1.9553	247

On the other hand, Table 3 provides a more direct comparison of the SS scores before and after fine-tuning, highlighting the notable change across categories.

The fine-tuned model shows a shift from anti-stereotype bias to a slight stereotype bias, with the overall SS score moving from 0.4915 to 0.5584. While this overshoots the ideal neutral point of 0.5, it demonstrates the effectiveness of the mitigation approach in modifying the model's associations.

Table 3 Comparison of SS Scores Before and After Fine-tuning

Category	Pre-FT SS	Post-FT SS	Change
Overall	0.4915	0.5584	+0.0669
Gender	0.4884	0.5624	+0.0740
Profession	0.4923	0.5582	+0.0659
Race	0.4920	0.5573	+0.0653
Religion	0.4863	0.5599	+0.0736

We can see the SS score comparison across all categories in Figure 4, providing a clear visualization of the shift in bias direction.

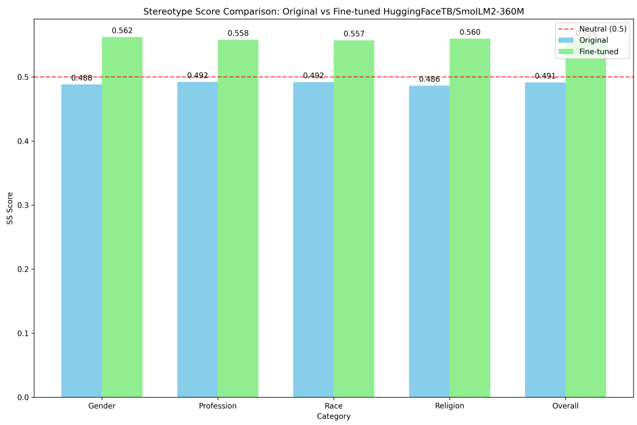

Fig. 4 Comparison of Stereotype Scores before and after fine-tuning

The most notable improvements occurred in the gender and religion categories, which were previously the most biased domains. The shift in bias direction from anti-stereotype to slight stereotype suggests that fine-tuning on anti-stereotypical examples may have over-corrected the model's preferences.

The log probability scores for both stereotypical and anti-stereotypical completions exhibited a significant improvement, indicating that the fine-tuning process enhanced the model's overall language modeling capabilities while adjusting its biases. For instance, the stereotype scores demonstrated a notable improvement, shifting from -4.5414 to -2.7129 for gender and from -4.4585 to -2.5029 for religion.

Figure 5 presents a radar chart comparing bias improvement across all domains, with the fine-tuned model shown to have more balanced scores across categories.

The post-fine-tuning bias distribution in Figure 6 shows a clear shift toward more balanced predictions, with a more concentrated distribution of scores around the neutral point.

The heatmap in Figure 7 further illustrates the more balanced association patterns in the fine-tuned model, with reduced extreme preferences across all categories.

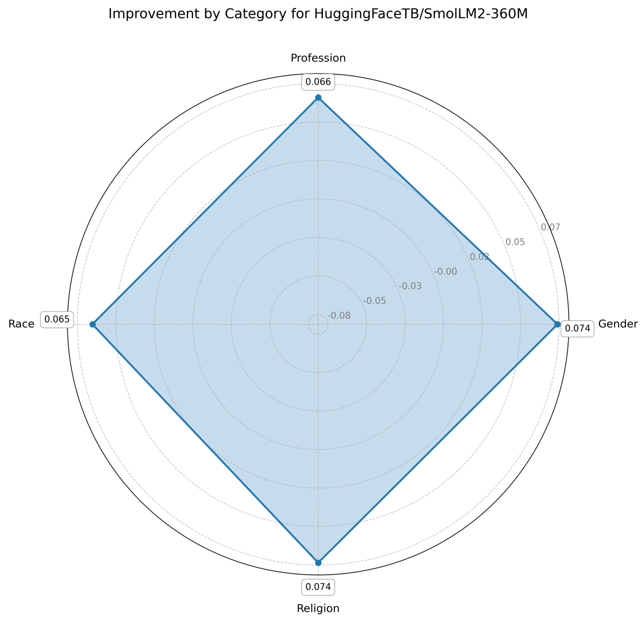

Fig. 5 Radar chart showing bias improvement across domains after fine-tuning

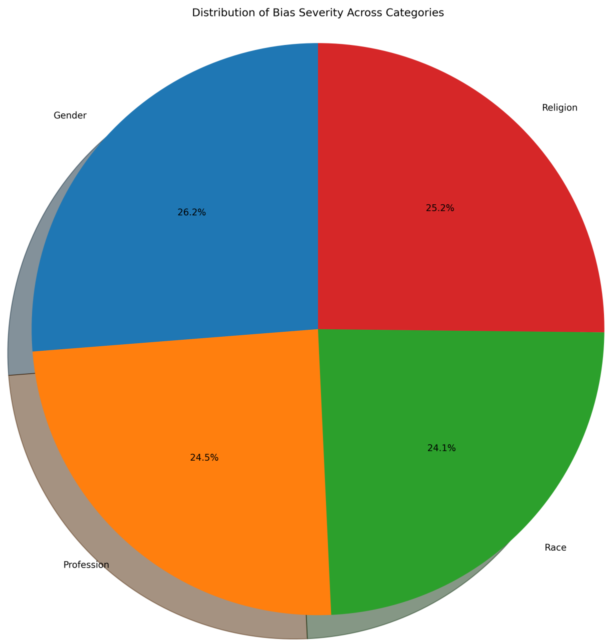

Fig. 6 Bias distribution after fine-tuning

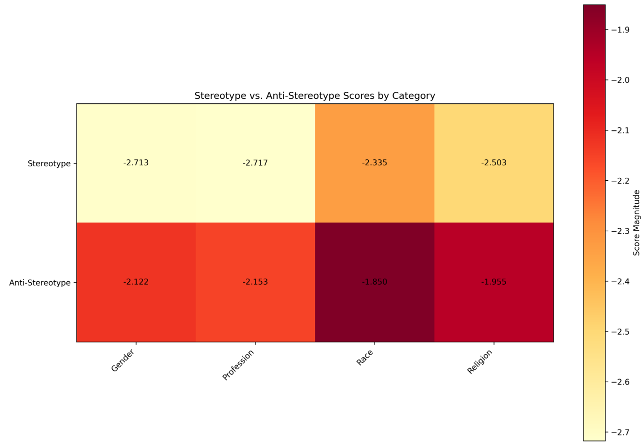

Fig. 7 Heatmap of stereotype vs. anti-stereotype associations after fine-tuning

The improvement in bias metrics came with no significant degradation in the model's language modeling performance, suggesting that the approach successfully balances fairness and utility.

5 Conclusion

This paper demonstrates both the persistence of social biases in lightweight transformer models and the effectiveness of efficient mitigation techniques. Through systematic evaluation and targeted fine-tuning, significant bias reduction was achieved in the SmolLM2-360M model across gender, profession, race, and religion domains.

The research reveals that social biases manifest differently across domains, with religion and gender showing the strongest bias in the target models, suggesting that bias mitigation strategies should prioritize these categories. Fine-tuning on counter-stereotypical examples proves to be an effective and computationally efficient approach to bias mitigation[6], with just three epochs of training producing significant changes in bias metrics. However, the results also reveal a tendency toward over-correction, with the model shifting from anti-stereotype bias to slight stereotype bias, highlighting the need for careful calibration of mitigation techniques to achieve true neutrality.

Importantly, the approach improved fairness without sacrificing language modeling performance, demonstrating that the goals of capability and fairness are not inherently in conflict, successfully balancing efficiency with meaningful bias reduction.

The findings contribute to the broader understanding of bias in language models and provide a practical framework for measuring and mitigating social biases in lightweight transformer architectures. By continuing to improve both measurement and mitigation techniques, progress can be made toward language models that maintain their utility while significantly reducing harmful stereotypes and biases. This advancement is essential as these technologies become increasingly embedded in critical systems that impact people's lives and opportunities.

References

[1] Yang, Y., Duan, H., Abbasi, A., Lalor, J.P., Tam, K.Y.: Bias A-head? Analyzing Bias in Transformer-Based Language Model Attention Heads (2024). https://arxiv.org/abs/2311.10395

[2] Nemani, P., Joel, Y.D., Vijay, P., Liza, F.F.: Gender Bias in Transformer Models: A comprehensive survey (2023). https://arxiv.org/abs/2306.10530

[3] Prakash, N., Lee, R.K.-W.: Layered bias: Interpreting bias in pretrained large language models. In: Belinkov, Y., Hao, S., Jumelet, J., Kim, N., McCarthy, A., Mohebbi, H. (eds.) Proceedings of the 6th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP, pp. 284–295. Association for Computational Linguistics, Singapore (2023). https://doi.org/10.18653/v1/2023.blackboxnlp-1.22 . https://aclanthology.org/2023.blackboxnlp-1.22/

[4] Nadeem, M., Bethke, A., Reddy, S.: StereoSet: Measuring stereotypical bias in pretrained language models (2020). https://arxiv.org/abs/2004.09456

[5] Allal, L.B., Lozhkov, A., Bakouch, E., Blázquez, G.M., Penedo, G., Tunstall, L., Marafioti, A., Kydlíček, H., Lajarín, A.P., Srivastav, V., Lochner, J., Fahlgren, C., Nguyen, X.-S., Fourrier, C., Burtenshaw, B., Larcher, H., Zhao, H., Zakka, C., Morlon, M., Raffel, C., Werra, L., Wolf, T.: SmolLM2: When Smol Goes Big – Data-Centric Training of a Small Language Model (2025). https://arxiv.org/abs/2502.02737

[6] Zhang, Y., Zhou, F.: Bias Mitigation in Fine-tuning Pre-trained Models for Enhanced Fairness and Efficiency (2024). https://arxiv.org/abs/2403.00625